

ENERGY-GUIDED DIFFUSION POLICY LIMITS: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

We study diffusion and energy-based robot policies. The motivating bet is: Identify when energy guidance changes reachable behavior rather than only re-ranking samples.. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For diffusion and energy-based robot policies, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2012; 2023; 2026a;b;c;d). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

3 MECHANISM

Given state s and action a , the model returns an observed next-state prediction plus a finite atlas $B(s, a)$ of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

Variant	Split	Success	Tail risk
observed_only_wam	nominal	0.555	0.445
observed_only_wam	hidden_mode_shift	0.185	0.815
augmentation_as_data_wam	nominal	0.570	0.430
augmentation_as_data_wam	hidden_mode_shift	0.247	0.752
scalar_uncertainty_wam	nominal	0.655	0.345
scalar_uncertainty_wam	hidden_mode_shift	0.335	0.665
proposed_branch_mechanism	nominal	0.748	0.252
proposed_branch_mechanism	hidden_mode_shift	0.588	0.412

5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

6 CONCLUSION

Energy-Guided Diffusion Policy Limits argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

REFERENCES

- Muscle coordination is habitual rather than optimal, 2012. <https://doi.org/10.1523/jneurosci.5792-11.2012>.
- Contrastive energy prediction for exact energy-guided diffusion sampling in offline reinforcement learning, 2023. <https://doi.org/10.48550/arxiv.2304.12824>.
- Prompt-guided large language model for estimating the state of charge of containerized lithium-ion battery energy storage systems, 2026a. <https://doi.org/10.2139/ssrn.6353339>.
- Dptg: diffusion policy with tactile feasibility guidance, 2026b. <https://doi.org/10.3389/frobt.2026.1851102>.
- Guided discovery of new behaviors using diffusion policies, 2026c. <http://arxiv.org/abs/2606.08743v1>.
- Progressvla: Progress-guided diffusion policy for vision-language robotic manipulation, 2026d. <https://openalex.org/W7148177967>.